A Practical Guide to Multi-armed bandit

TOPIC -- MULTI-ARMED BANDIT

-- 01 --

Guha, S.; Munagala, K.; Shi, P. (2010), "Approximation algorithms for restless bandit problems", Journal of the ACM, 58: 1–50, arXiv:0711.3861, doi:10.1145/1870103.1870106, S2CID 1654066 Dayanik, S.; Powell, W.; Yamazaki, K. (2008), "Index policies for discounted bandit problems with availability constraints", Advances in Applied Probability, 40 (2): 377–400, doi:10.1239/aap/1214950209. Powell, Warren B. (2007), "Chapter 10", Approximate Dynamic Programming: Solving the Curses of Dimensionality, New York: John Wiley and Sons, ISBN 978-0-470-17155-4. Robbins, H. (1952), "Some aspects of the sequential design of experiments", Bulletin of the American Mathematical Society, 58 (5): 527–535, Bibcode:1952BAMaS..58..527R, doi:10.1090/S0002-9904-1952-09620-8. Sutton, Richard; Barto, Andrew (1998), Reinforcement Learning, MIT Press, ISBN 978-0-262-19398-6, archived from the original on 2013-12-11. Allesiardo, Robin (2014), "A Neural Networks Committee for the Contextual Bandit Problem", Neural Information Processing – 21st International Conference, ICONIP 2014, Malaisia, November 03-06,2014, Proceedings, Lecture Notes in Computer Science, vol. 8834, Springer, pp. 374–381, arXiv:1409.8191, doi:10.1007/978-3-319-12637-1_47, ISBN 978-3-319-12636-4, S2CID 14155718. Weber, Richard (1992), "On the Gittins index for multiarmed bandits", Annals of Applied Probability, 2 (4): 1024–1033, doi:10.1214/aoap/1177005588, JSTOR 2959678. Katehakis, M.; C. Derman (1986), "Computing optimal sequential allocation rules in clinical trials", Adaptive statistical procedures and related topics, Institute of Mathematical Statistics Lecture Notes - Monograph Series, vol. 8, pp. 29–39, doi:10.1214/lnms/1215540286, ISBN 978-0-940600-09-6, JSTOR 4355518. Katehakis, Michael N.; Veinott, Jr., Arthur F. (1987), "The multi-armed bandit problem: decomposition and computation", Mathematics of Operations Research, 12 (2): 262–268, doi:10.1287/moor.12.2.262, JSTOR 3689689, S2CID 656323

-- 02 --

Semi-uniform strategies Semi-uniform strategies were the earliest (and simplest) strategies discovered to approximately solve the bandit problem. All those strategies have in common a greedy behavior where the best lever (based on previous observations) is always pulled except when a (uniformly) random action is taken.

-- 03 --

Exp3 Source: EXP3 is a popular algorithm for adversarial multiarmed bandits, suggested and analyzed in this setting by Auer et al. [2002b]. Recently there was an increased interest in the performance of this algorithm in the stochastic setting, due to its new applications to stochastic multi-armed bandits with side information [Seldin et al., 2011] and to multi-armed bandits in the mixed stochastic-adversarial setting [Bubeck and Slivkins, 2012]. The paper presented an empirical evaluation and improved analysis of the performance of the EXP3 algorithm in the stochastic setting, as well as a modification of the EXP3 algorithm capable of achieving "logarithmic" regret in stochastic environment.

-- 04 --

Add noise to each arm and pull the one with the highest value 3. Update value: R I (t) (t + 1) = R I (t) (t) + x I (t) (t) {\displaystyle R_{I(t)}(t+1)=R_{I(t)}(t)+x_{I(t)}(t)}

-- 05 --

UCB-ALP algorithm: The framework of UCB-ALP is shown in the right figure. UCB-ALP is a simple algorithm that combines the UCB method with an Adaptive Linear Programming (ALP) algorithm, and can be easily deployed in practical systems. It is the first work that show how to achieve logarithmic regret in constrained contextual bandits. Although is devoted to a special case with single budget constraint and fixed cost, the results shed light on the design and analysis of algorithms for more general CCB problems.

-- 06 --

An important variation of the classical regret minimization problem in multi-armed bandits is best arm identification (BAI), also known as pure exploration. This problem is crucial in various applications, including clinical trials, adaptive routing, recommendation systems, and A/B testing. In BAI, the objective is to identify the arm having the highest expected reward. An algorithm in this setting is characterized by a sampling rule, a decision rule, and a stopping rule, described as follows:

-- 07 --

For each t = 1, 2, ..., T 1. Set p i (t) = (1 − γ) ω i (t) ∑ j = 1 K ω j (t) + γ K {\displaystyle p_{i}(t)=(1-\gamma){\frac {\omega _{i}(t)}{\sum _{j=1}^{K}\omega _{j}(t)}}+{\frac {\gamma }{K}}} i = 1 , . . . , K {\displaystyle i=1,...,K}

-- 08 --

ρ π (T) = ∑ t = 1 T μ t ∗ − E π μ [∑ t = 1 T r t] = D (T) − E π μ [∑ t = 1 T r t] . {\displaystyle \rho ^{\pi }(T)=\sum _{t=1}^{T}{\mu _{t}^{*}}-\mathbb {E} _{\pi }^{\mu }\left[\sum _{t=1}^{T}{r_{t}}\right]={\mathcal {D}}(T)-\mathbb {E} _{\pi }^{\mu }\left[\sum _{t=1}^{T}{r_{t}}\right].}

-- 09 --

Collaborative bandit Approaches using multiple bandits that cooperate sharing knowledge in order to better optimize their performance started in 2013 with "A Gang of Bandits", an algorithm relying on a similarity graph between the different bandit problems to share knowledge. The need of a similarity graph was removed in 2014 by the work on the CLUB algorithm. Following this work, several other researchers created algorithms to learn multiple models at the same time under bandit feedback. For example, COFIBA was introduced by Li and Karatzoglou and Gentile (SIGIR 2016), where the classical collaborative filtering, and content-based filtering methods try to learn a static recommendation model given training data.

-- 10 --

Epsilon-greedy strategy: The best lever is selected for a proportion 1 − ■ {\displaystyle 1-\epsilon } of the trials, and a lever is selected at random (with uniform probability) for a proportion ■ {\displaystyle \epsilon } . A typical parameter value might be ■ = 0.1 {\displaystyle \epsilon =0.1} , but this can vary widely depending on circumstances and predilections. Epsilon-first strategy: A pure exploration phase is followed by a pure exploitation phase. For N {\displaystyle N} trials in total, the exploration phase occupies ■ N {\displaystyle \epsilon N} trials and the exploitation phase (1 − ■) N {\displaystyle (1-\epsilon)N} trials. During the exploration phase, a lever is randomly selected (with uniform probability); during the exploitation phase, the best lever is always selected. Epsilon-decreasing strategy: Similar to the epsilon-greedy strategy, except that the value of ■ {\displaystyle \epsilon } decreases as the experiment progresses, resulting in highly explorative behaviour at the start and highly exploitative behaviour at the finish. Adaptive epsilon-greedy strategy based on value differences (VDBE): Similar to the epsilon-decreasing strategy, except that epsilon is reduced on basis of the learning progress instead of manual tuning (Tokic, 2010). High fluctuations in the value estimates lead to a high epsilon (high exploration, low exploitation); low fluctuations to a low epsilon (low exploration, high exploitation). Further improvements can be achieved by a softmax-weighted action selection in case of exploratory actions (Tokic & Palm, 2011). Adaptive epsilon-greedy strategy based on Bayesian ensembles

(Epsilon-BMC): An adaptive epsilon adaptation strategy for reinforcement learning similar to VBDE, with monotone convergence guarantees. In this framework, the epsilon parameter is viewed as the expectation of a posterior distribution weighting a greedy agent (that fully trusts the learned reward) and uniform learning agent (that distrusts the learned reward). This posterior is approximated using a suitable Beta distribution under the assumption of normality of observed rewards. In order to address the possible risk of decreasing epsilon too quickly, uncertainty in the variance of the learned reward is also modeled and updated using a normal-gamma model. (Gimelfarb et al., 2019).